OCTAL TO BINARY
&
BINARY TO OCTAL

CONVESRIONS

CONSEPT

1. Octal has a base of 8
2. Binary has a base of 2

$8 = 2^3$

every octal digit can be represented using 3 binary bits.

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

BINARY TO OCTAL RULE

1. Start grouping the binary digits from the right-hand side. Each group should contain exactly 3 bits.

2. If the leftmost group has fewer than 3 bits, simply add leading zeros.

3. After grouping, replace each binary group with its equivalent octal digit.

Example 1

$101101_2 = ?_8$

101 101

5 5

$101101_2 = 55_8$

Example 2

$11011_2 = ?_8$

0 1 1 0 1 1

3 3

$11011_2 = 33_8$

Example 3

$10110.10101_2 = ?_8$

0 10   110 . 101   01 0

2   6   5   2

$10110.10101_2 = 26.52_8$

PRACTICE QUESTIONS

$100101_2 = ?_8$
$111010101_2 = ?_8$
$1010111.101_2 = ?_8$
$11001.11101_2 = ?_8$
$111111000_2 = ?_8$

OCTAL TO BINARY CONVERSION

Example 1

$35_8 = ?_2$

$3 \rightarrow 011 \quad 5 \rightarrow 101$

011 101

$35_8 = 11101_2$

Example 2

$25.6_8 = (\quad)_2$

2   5 . 6

010   101 . 110

010101.110

$25.6_8 = 10101.110_2$

PRACTICE QUESTIONS

$6315_8 = ?_2$

$72.54_8 = ?_2$

thank you

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